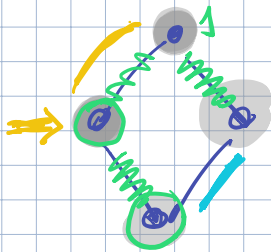
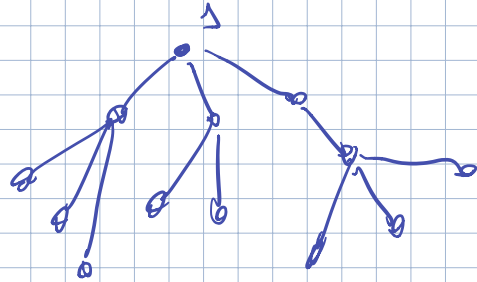
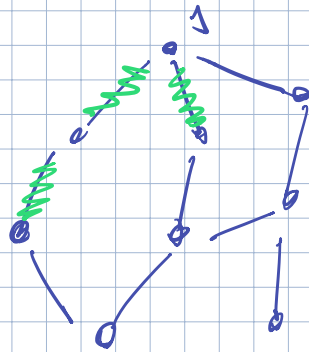


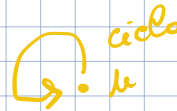
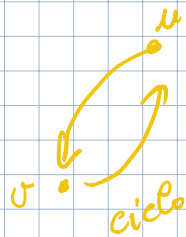
Es. Grafo non orient. ciclico



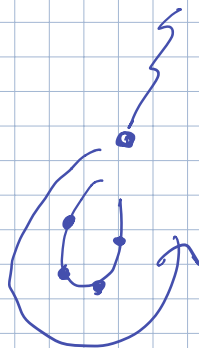
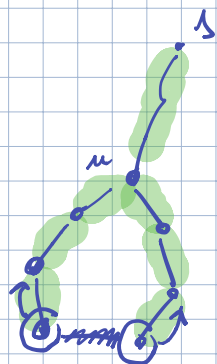
Archi da grigio
e non bianco

trattare gli archi
che tornano a $\pi[u]$

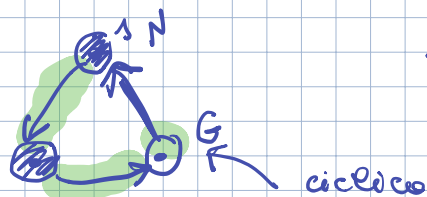
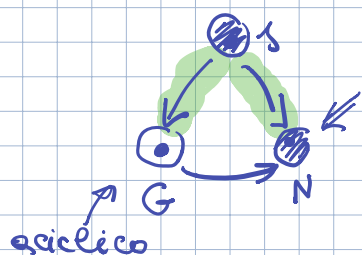
Funziona nei grafi orientati?



G non orientato ciclico?



G orientato



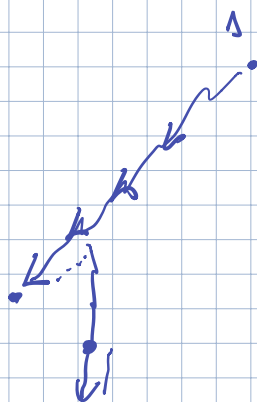
BFS

Non riesco a distinguerli con BFS



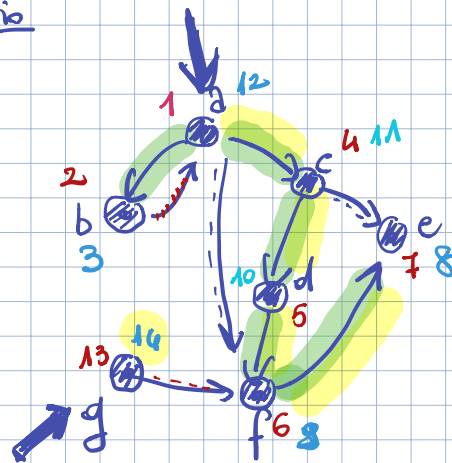
Visita DFS

Profondità



Seguo un cammino in profondità
Faccio Backtracking quando necessario

Esempio



$\sum$
 P: ea
 $\sum$
 Ricorsione

- Colori
- da visite riportate se sono rimasti nodi bianchi
- π albero di visite DFS
- tempi inizio e fine visite
 $i \rightarrow f$

DFS (G) {

for (each $v \in V$) {
 color[v] $\leftarrow$ B
 $\pi[v] \leftarrow NIL$

$\sum$
 $\Theta(|V|)$

}

TIME $\leftarrow$ 1

for (each $v \in V$) {

 if (color[v] = B) {

 TIME $\leftarrow$ DFS-visit (G, v, TIME) $\leftarrow$

 }

}

}

DFS-visit (G, v, TIME) {

①(u) { color [v] ← Grigio
i [v] ← TIME
TIME ← TIME + 1

{ for (each $u \in \text{Adj}[v]$) {
if (color [u] = B) {
π [u] ← v
→ TIME ← DFS-visit (G, u, TIME)
}
}
}

②(H) { f [v] ← TIME
TIME ← TIME + 1
color [v] ← N
return TIME

}

Complessità

$$\Theta(n) + \Theta(n) + \Theta\left(\sum_{v \in V} |\text{Adj}[v]| \right)$$

operazioni
①(1)

Matrici
di Adj Liste
di Adj

Matrici di Adj

$$\sum_u (|V|) + \sum_{u \in V} (|V|) = \sum_u (|V|^2)$$

Liste di Adj

$$\sum_u (|V|) + \sum_{u \in V} (|L_G[u]|) = \sum_u (|E|)$$

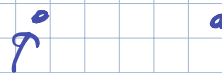
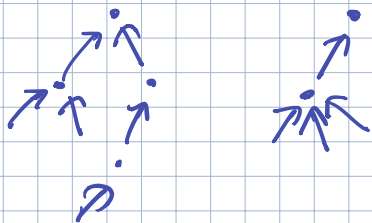
$$= \sum_u (|V| + |E|)$$

Prop.



$$V = \{1, 2, \dots, |V|\}$$

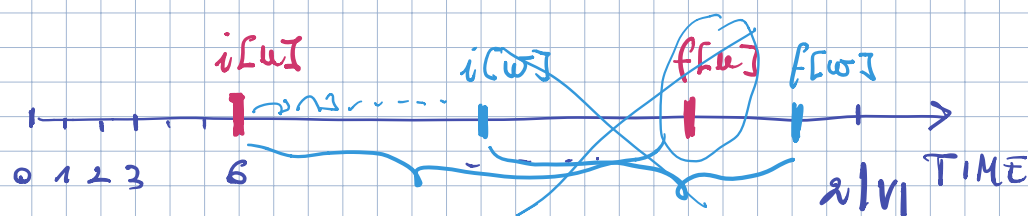
NIZ alla fine rappresentano
i nodi sui quali DFS-visit
è partita da DFS



Foreste
di Alberi

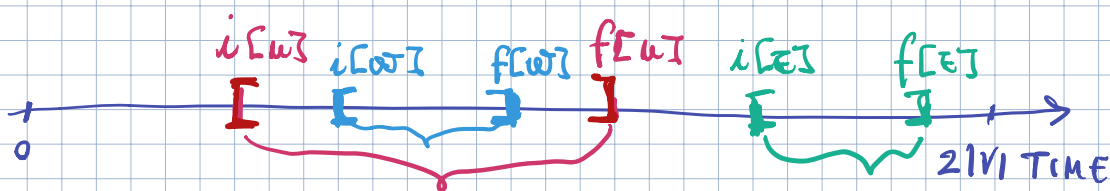
Nel caso di Grafi non orientati
 non ha un unico albero di visita se
 G è connesso

Prop. i f



$$i[u] < f[u]$$

Non può
 succedere



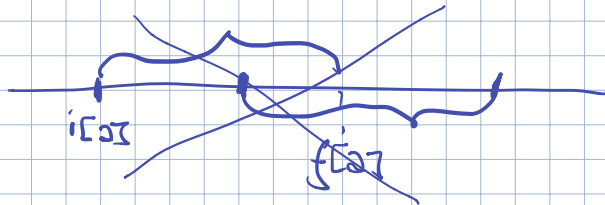
Teorema delle Parentesi

Per ogni $a, b \in V$

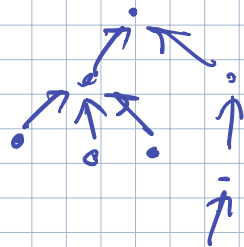
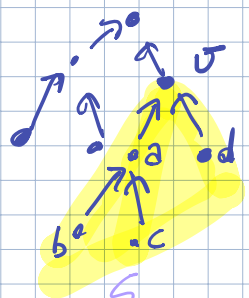
- $[i[a], f[a]] \cap [i[b], f[b]] = \emptyset$ opp.

- $[i[a], f[a]] \subseteq [i[b], f[b]]$ oppure
- $[i[b], f[b]] \subseteq [i[a], f[a]]$

Sicuramente non si verifica

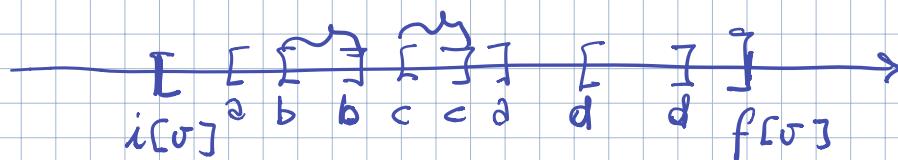


Quali sono nelle foreste di alberi
DFS i discendenti di un nodo v ?



lemano
i tempi

Scoperti
grazie a v
Discendenti di v

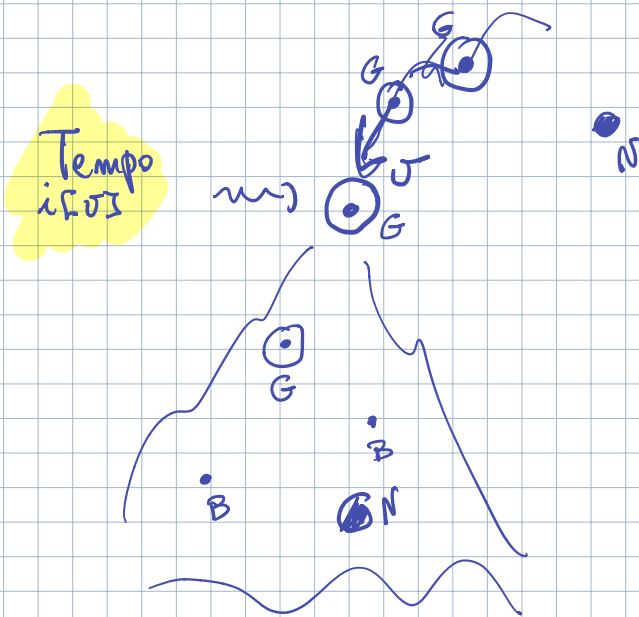


Teo

u è discendente di v nelle foreste

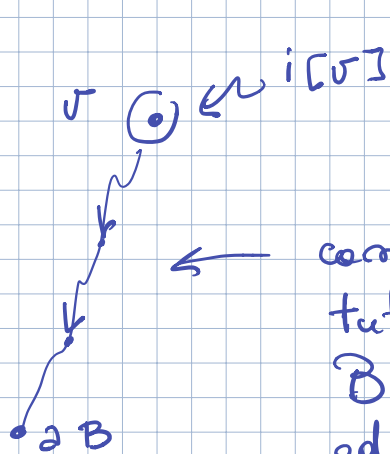
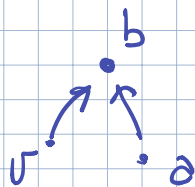
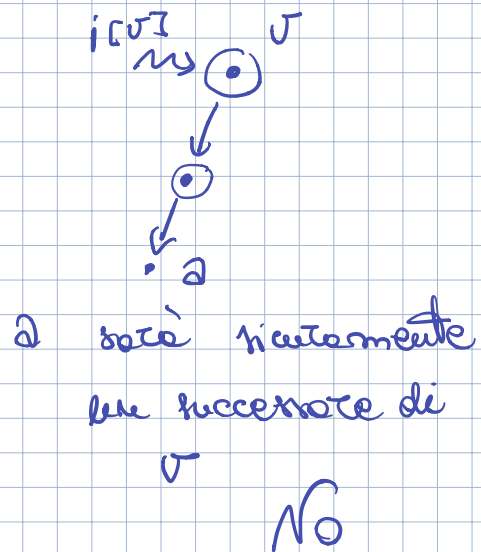
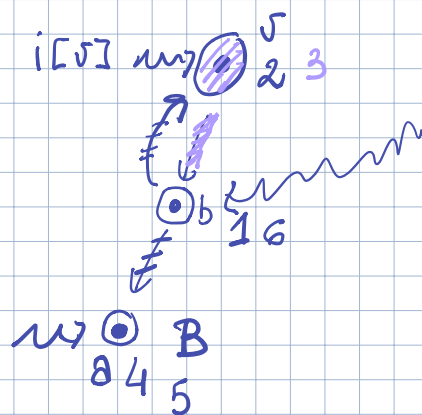
DFS se $[i[u], f[u]] \subseteq [i[v], f[v]]$

Quali sono nelle foreste di alberi
DFS i discendenti di un nodo v ?



Usando i colori
Durante
l'esecuzione
di DFS

Tutti i nodi
che sono bianchi
e raggiungibili
da v
verranno scoperti
grazie a v ?



Teorema del cammino Bianco

cammino
tutto di nodi
Bianchi da v
ad a al tempo

$i[v]$



a sarà discendente
di v nella
Foresta DFS